

ECSP3000 Project Brief

A Reproducible Evaluation of Linux Rootkit Detectors: A Testbed and Proof-of-Concept Rootkit

Elliott Jelbert

ejlg24@soton.ac.uk

Supervisor: Dr. BooJoong Kang

1. Project Aims

Current research into the evaluation of rootkit detectors rely on bespoke evaluation methodologies which cannot be reliably or easily reproduced by independent parties. Furthermore, as the Linux kernel is updated, the mechanisms that rootkits exploit change, as do the techniques used by detectors. This creates a need for a consistent way to evaluate rootkit detector tools continuously as new rootkit techniques are developed and new detection tools are produced.

This project aims to fulfill this need by producing a tool that provides developers and researchers with an automated and reproducible tool for evaluating rootkit detectors against rootkits. It also aims to conduct an in-depth analysis of the effectiveness of on-host detection techniques on the modern Linux kernel via this testbed and a modern proof-of-concept rootkit.

2. Project Objectives

The core goal for this project is to develop a reproducible & automated testbed for linux rootkit detectors, delivering an in depth analysis of the effectiveness of on-host rootkit detectors on the modern Linux kernel.

Research Goals

- The testbed produces **reproducible** and **accurate** results in a **timely** manner.
- The testbed must allow developers to easily add new rootkit detectors or rootkits into the system
- As a stretch, design & implement a proof-of-concept rootkit for the 7.x Linux kernel series, to be used in my evaluation.
- To combine the testbed & a set of modern rootkits & rootkit detectors to produce an evaluation of the current effectiveness of on-host rootkit detection tools on key metrics including detection rate, false positives and false negatives.

3. Limitations & Scope

This project aims to investigate on-host rootkit detection techniques only. An on-host detection tool is any tool that is installed & ran locally on the potentially infected machine which then reports back to the user the presence of a rootkit on the system. This excludes memory forensic techniques, hypervisor-based techniques or any other out-of-band method from the scope. Similarly, only kernel-space & user-space rootkits will be used in testing. Additionally these rootkits must be public & open-source and not employ any anti-sandboxing measures which may impact results due to the virtualised environment used in testing.

My research will be for the Linux kernel versions 6.x and above, ensuring my project stays relevant by targeting a modern platform that is currently understudied in regards to rootkit detection.